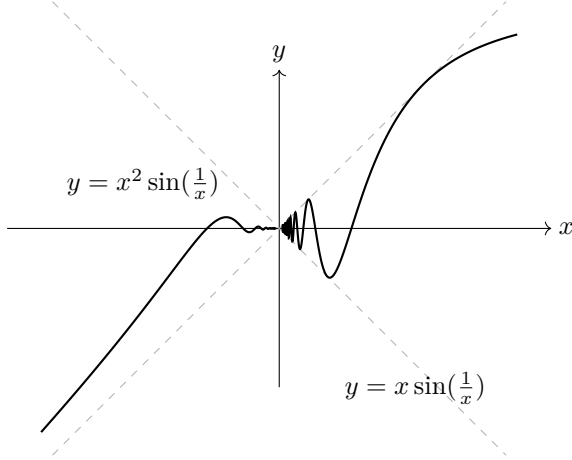


0.1 The function $x^a \sin(|x|^{-c})$

Definition 0.1.1. Define a map as: for $a \in \mathbb{R}$ and $c > 0$,

$$f_a^c : [-1, 1] \rightarrow \mathbb{R} : x \mapsto \begin{cases} 0 & \text{if } x = 0 \\ x^a \sin(|x|^{-c}) & \text{if } x \neq 0 \end{cases}$$



Equivalent Condition Table	f_a^c	f'
Continuous	$a > 0$	$a > 1 + c$
Bounded	$a \geq 0$	$a \geq 1 + c$
Differentiable at $x = 0$	$a > 1$	-
Bounded Variation	$a \geq c$	-

Lemma 0.1.2. The derivative of $f_a^c(x)$ is given by:

$$\frac{d}{dx} f_a^c(x) = \begin{cases} \lim_{t \rightarrow 0} t^{a-1} \sin\left(\frac{1}{|t|^c}\right) & \text{if } x = 0 \text{ (provided the limit exists)} \\ ax^{a-1} \sin\left(\frac{1}{x^c}\right) - cx^{a-c-1} \cos\left(\frac{1}{x^c}\right) & \text{if } x > 0 \end{cases}$$

0.2 Typical Examples

Proposition 0.2.1. f_1^1 is continuous, **Not** differentiable at $x = 0$, **Not** of bounded variation, and does **Not** have a bounded derivative.

Proof. Since $\left| x \sin\left(\frac{1}{x}\right) \right| \leq |x| \rightarrow 0$ as $x \rightarrow 0$, f_1^1 is clearly continuous.

Meanwhile, $f'(0)$ exists if and only if $\lim_{t \rightarrow 0} \frac{f(t) - f(0)}{t} = \lim_{t \rightarrow 0} \sin\left(\frac{1}{t}\right)$ exists.

However, this limit does not exist, so f_1^1 is not differentiable at $x = 0$. To show that f_1^1 is not of bounded variation, we construct a partition P_n as:

$$P_n \stackrel{\text{def}}{=} \left\{ 0, \frac{2}{n\pi}, \frac{2}{(n-1)\pi}, \dots, \frac{2}{2\pi}, \frac{2}{\pi}, 1 \right\}$$

Then, the variation satisfies:

$$V(f_1^1, P_{2n+1}) \geq \sum_{k=0}^{n-1} \frac{2}{(2k+1)\pi} = \frac{2}{\pi} \sum_{k=0}^{n-1} \frac{1}{2k+1} \rightarrow \infty \text{ as } n \rightarrow \infty.$$

Therefore, f_1^1 is not of bounded variation. Similarly, f_n^n ($n \in \mathbb{N}$) are not of bounded variation. □

But, in this way, we cannot determine whether $f_2^1(x) = x^2 \sin\left(\frac{1}{x}\right)$ is bounded variation or not.

Proposition 0.2.2. If $f : [a, b] \rightarrow \mathbb{R}$ is differentiable and the derivative f' is bounded, then f is of bounded variation.

Proof. For any partition P on $[0, 1]$, by the Mean Value Theorem, there exists $t_i \in (x_{i-1}, x_i)$ such that

$$V(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i \leq \sum_{i=1}^n \sup |f'| \cdot \Delta x_i = \sup |f'| \cdot 1$$

Since P was arbitrary, f is bounded variation. □

Corollary 0.2.3. f_2^1 is of bounded variation on $[-1, 1]$.

Proposition 0.2.4. f_a^c is bounded variation on $[0, 1]$ if and only if $a > c$

Proof. Suppose that $a \leq c$. Then, similarly in the case of f_1^1 , take a partition P_n on $[0, 1]$ as:

$$P_n \stackrel{\text{def}}{=} \left\{ 0, \sqrt[n]{\frac{c}{n\pi}}, \sqrt[n]{\frac{c}{(n-1)\pi}}, \dots, \sqrt[n]{\frac{c}{2\pi}}, \sqrt[n]{\frac{c}{\pi}}, 1 \right\}$$

Then,

$$V(f_a^c, P_{2n+1}) \geq \sum_{k=0}^{n-1} \left(\frac{2}{(2k+1)\pi} \right)^{\frac{a}{c}} \rightarrow \infty \text{ as } n \rightarrow \infty$$

because $a \leq c$ implies $\frac{a}{c} \leq 1$. (In case of $a > c$ will be continue) □